

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4, suppress=True)

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Dispositivo: {device}')
```

Dispositivo: cuda

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints
NAMING_META = {
    'variante': 'topk',
    'tecnica': 'batch-topk',
    'capa': 6,
    'juegos': 50000,
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../../experiments'),
    'extras': {
        'expansion': 16,
        'k': 81,
        'epochs': 1000,
        'patience': 20,
        'batch_size_tokens': 1024,
    }
}

print('\nMetadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')
```

Metadata de naming:

```
variante: topk
tecnica: batch-topk
capa: 6
juegos: 50000
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 16, 'k': 81, 'epochs': 1000, 'patience': 20, 'batch_size_tokens': 1024}
```

```
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
```

```

    NAMING_META['extras']
) if NAMING_META.get('extras') else None

# get_plots_dir crea el directorio automáticamente
plots_dir = Path(get_plots_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
))

print(f'extras_id: {extras_id}')
print(f'plots_dir: {plots_dir}')

```

Nuevo extras registrado: ex2 → {'expansion': 16, 'k': 81, 'epochs': 1000, 'patience': 20, 'extras_id': ex2
plots_dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_topk\50000games\ex2\plots

3. Configuración del SAE con SAELens

Configuración usando LanguageModelSAERunnerConfig de SAELens con architecture='batchtopk'.

Arquitectura BatchTopK: - **BatchTopK activation:** calcula un threshold global por batch para que el **promedio** de features activas por muestra sea **k** (L0 promedio k , pero varía por muestra) - **Sin coeficiente de sparsidad:** la sparsity se controla directamente con **k** - **Loss = MSE únicamente** (sin penalización L1)

Parámetro	L1 (standard)	TopK	BatchTopK
Sparsity	sparsity_coef (soft)	k (hard, por muestra)	k (promedio del batch)
L0	Variable	Fijo = k por muestra	Variable por muestra, promedio k
Loss	$\text{MSE} + \lambda \cdot \text{L1}$	MSE	MSE

Nota sobre el runner: LanguageModelSAETrainingRunner(cfg).run() requiere que Othello-GPT esté cargado vía TransformerLens. Las celdas siguientes entrenan

el SAE directamente sobre los archivos .npy pre-extraídos usando la arquitectura BatchTopK definida en el config.

```
NUM_EPOCHS = NAMING_META['extras']['epochs']

cfg = {
    'architecture': 'batchtopk',
    'd_in': 512,
    'd_sae': 8192,
    'k': 81,
    'lr': 1e-3,
    'train_batch_size_tokens': 1024,
    'hook_name': 'blocks.5.hook_resid_post',
}

cfg_summary = pd.DataFrame({
    'Parámetro SAELens': ['architecture', 'd_in', 'd_sae', 'k (L0 promedio)', 'hook_name', 'lr'],
    'Valor': [cfg['architecture'], cfg['d_in'], cfg['d_sae'], cfg['k'],
              cfg['hook_name'], cfg['lr']]
})

print('Configuración SAELens:')
print(cfg_summary.to_string(index=False))
```

Configuración SAELens:

Parámetro SAELens	Valor
architecture	batchtopk
d_in	512
d_sae	8192
k (L0 promedio)	81
hook_name	blocks.5.hook_resid_post
lr	0.001

4. Dataset de Activaciones (Train/Validation Split 80-20%)

Activaciones pre-extraídas con sae/activations/run_extraction.py desde la capa blocks.5.hook_resid_post.

```
class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""
    def __init__(self, activations_path):
        self.activations = np.load(activations_path)
```

```

def __len__(self):
    return len(self.activations)
def __getitem__(self, idx):
    return torch.tensor(self.activations[idx], dtype=torch.float32)

activations_file = Path(f"../../activations/data/layer{NAMING_META['capa']}_{NAMING_META['id']}")
full_dataset = ActivationsDataset(activations_file)

# Split 80-20 reproducible
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset, [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

train_loader = DataLoader(train_dataset, batch_size=cfg["train_batch_size_tokens"])
val_loader = DataLoader(val_dataset, batch_size=cfg["train_batch_size_tokens"])

dataset_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}'],
    'Shape': [str(full_dataset.activations.shape), '', ''],
})
print(dataset_info.to_string(index=False))

```

```

Conjunto  Muestras      Shape
Train 2,360,000 (2950000, 512)
Validación  590,000
Total 2,950,000

```

5. Modelo: BatchTopK SAE

Implementación de la arquitectura BatchTopK de SAELens.

Diferencia clave vs ReLU + L1 y vs TopK estándar: - El encoder proyecta a `d_sae=8192` dimensiones (pre-activación) - La activación **BatchTopK** calcula un threshold global del batch (k-ésimo valor promediado); cada muestra puede tener más o menos de `k` features activas - L0 es **variable por muestra**, pero el **promedio del batch** `k` (soft constraint a nivel de batch) - No hay `sparsity_coef` ni cosine annealing - El decoder se normaliza a norma unitaria después de cada paso del optimizador

```

class BatchTopKSAE(nn.Module):
    """
    Sparse Autoencoder con activación BatchTopK.
    Equivalente a SAELens architecture='batchtopk'.

    Encoder      : d_in -> d_sae (lineal + pre-bias centering)
    Activación    : BatchTopK (mantiene el promedio de k features activas por batch)
    Decoder       : d_sae -> d_in (columnas con norma unitaria)
    """
    def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
        super().__init__()
        self.d_in = d_in
        self.d_sae = d_sae
        self.k = k
        self.k_aux = k_aux

        self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
        self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
        self.b_dec = nn.Parameter(torch.zeros(d_in))

        nn.init.kaiming_uniform_(self.W_enc)
        nn.init.kaiming_uniform_(self.W_dec)
        self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
        self.register_buffer('num_batches_not_active', torch.zeros(d_sae))
        self.n_batches_to_dead = 10 # latente muerto si inactivo por 10 batches seguidos

    def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
        """Proyecta a espacio latente con activación BatchTopK"""
        x_cent = x - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        return self._topk_activation(pre_acts, threshold=threshold)

    def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
        """BatchTopK: selecciona exactamente k * batch_size activaciones usando scatter."""
        if threshold is None:
            acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
            return (
                torch.zeros_like(x.flatten())
                .scatter(-1, acts_topk.indices, acts_topk.values)
                .reshape(x.shape)
            )
        return torch.where(x > threshold, x, torch.zeros_like(x))

```

```

def decode(self, hidden: torch.Tensor) -> torch.Tensor:
    return hidden @ self.W_dec + self.b_dec

def preprocess_input(self, x: torch.Tensor):
    x_mean = x.mean(dim=-1, keepdim=True)
    x = x - x_mean
    x_std = x.std(dim=-1, keepdim=True)
    x = x / (x_std + 1e-5)
    return x, x_mean, x_std

def postprocess_output(self, x_reconstruct: torch.Tensor,
                       x_mean: torch.Tensor, x_std: torch.Tensor) -> torch.Tensor:
    return x_reconstruct * x_std + x_mean

def forward(self, x: torch.Tensor, threshold=None):
    x_norm, x_mean, x_std = self.preprocess_input(x)
    x_cent = x_norm - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    hidden = self._topk_activation(pre_acts, threshold=threshold)
    recon = self.postprocess_output(self.decode(hidden), x_mean, x_std)
    return recon, hidden, pre_acts

def auxiliary_loss(self, x: torch.Tensor, x_reconstruct: torch.Tensor, acts: torch.Tensor):
    """L_aux del repo oficial: reconstruye el residual con top-k_aux dead latents."""
    dead_features = self.num_batches_not_active >= self.n_batches_to_dead
    if dead_features.sum() == 0:
        return torch.tensor(0.0, device=x.device)
    residual = x.float() - x_reconstruct.float()
    acts_topk_aux = torch.topk(
        acts[:, dead_features],
        min(self.k_aux, int(dead_features.sum().item())) ,
        dim=-1,
    )
    acts_aux = torch.zeros_like(acts[:, dead_features]).scatter(
        -1, acts_topk_aux.indices, acts_topk_aux.values
    )
    x_reconstruct_aux = acts_aux @ self.W_dec[dead_features]
    return (1/32) * (x_reconstruct_aux.float() - residual.float()).pow(2).mean()

@torch.no_grad()
def update_inactive_features(self, acts):
    self.num_batches_not_active += (acts.sum(0) == 0).float()

```

```

        self.num_batches_not_active[acts.sum(0) > 0] = 0

    @torch.no_grad()
    def estimate_threshold(self, loader, device, n_batches: int = 10) -> float:
        """Estima un threshold estable promediando sobre múltiples batches
        (independiente del batch actual - para usar en validación/inferencia)."""
        self.eval()
        thresholds = []
        for i, batch in enumerate(loader):
            if i >= n_batches:
                break
            x = batch.to(device)
            x_norm, _, _ = self.preprocess_input(x)
            x_cent = x_norm - self.b_dec
            pre_acts = F.relu(x_cent @ self.W_enc)
            mins = []
            for sample_idx in range(pre_acts.shape[0]):
                pos = pre_acts[sample_idx][pre_acts[sample_idx] > 0]
                if pos.numel() > 0:
                    mins.append(pos.min().item())
            t = float(np.mean(mins)) if mins else 0.0
            thresholds.append(t)
        return float(np.mean(thresholds))

    @torch.no_grad()
    def make_decoder_weights_and_grad_unit_norm(self):
        """Proyecta el gradiente del decoder y renormaliza - igual que el repo oficial."""
        W_dec_normed = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)
        W_dec_grad_proj = (self.W_dec.grad * W_dec_normed).sum(-1, keepdim=True) * W_dec_normed
        self.W_dec.grad -= W_dec_grad_proj
        self.W_dec.data = W_dec_normed

# Crear modelo desde parámetros del cfg SAELens
model = BatchTopKSAE(d_in=cfg['d_in'], d_sae=cfg['d_sae'], k=cfg['k']).to(device)
num_params = sum(p.numel() for p in model.parameters())

model_info = pd.DataFrame({
    'Componente': [
        'Input', 'Encoder (W_enc)', 'Activación', 'Hidden activos', 'Decoder (W_dec)', 'Output'
    ],
    'k_aux', 'Total params'
})

```



```

'Dimensión': [
    f'{cfg["d_in"]}',
    f'{cfg["d_in"]} -> {cfg["d_sae"]}',
    f'BatchTopK (k={cfg["k"]})',
    f'{cfg["d_sae"]} (solo {cfg["k"]} activos en promedio)',
    f'{cfg["d_sae"]} -> {cfg["d_in"]}',
    f'{cfg["d_in"]}',
    f'{model.k_aux}',
    f'{num_params:}'
]
})
print(model_info.to_string(index=False))

```

Componente	Dimensión
Input	512
Encoder (W_{enc})	512 -> 8192
Activación	BatchTopK (k=81)
Hidden activos	8192 (solo 81 activos en promedio)
Decoder (W_{dec})	8192 -> 512
Output	512
k_{aux}	512
Total params	8,389,120

6. Función de Pérdida BatchTopK

Loss = MSE(reconstrucción)

- **MSE**: error de reconstrucción (igual que en L1)
- **Sin L1 penalty**: la sparsity ya está controlada por el threshold global de BatchTopK
- No hay `sparsity_coef` ni annealing — el único hiperparámetro de sparsity es **k** (promedio del batch)

```

ALPHA_AUX = 1 / 32 # peso de la pérdida auxiliar (arXiv:2412.06410)

def loss_function(x, reconstruction, acts, model):
    """
    Pérdida BatchTopK: MSE + alpha * L_aux (arXiv:2412.06410).
    - MSE      : error de reconstrucción
    - L_aux    : reconstruye el residual con las top-k_aux features para
                  evitar features muertas (alpha = 1/32)
    """

```

```

Args:
    x: Activaciones originales
    reconstruction: Activaciones reconstruidas
    acts: Pre-activaciones del encoder
    model: Instancia de BatchTopKSAE
Returns:
    total_loss (scalar)
"""
mse = F.mse_loss(reconstruction, x)
l_aux = model.auxiliary_loss(x, reconstruction, acts)
return mse + l_aux

optimizer = torch.optim.Adam(model.parameters(), lr=cfg["lr"], betas=(0.9, 0.99))

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning rate', 'betas', 'k (L0 promedio k)', 'k_aux', 'alpha_aux'],
    'Valor': ['Adam', cfg["lr"], '(0.9, 0.99)', cfg["k"], model.k_aux, f'{ALPHA_AUX:.5f}']
})
print(optim_info.to_string(index=False))

```

	Parámetro	Valor
	Optimizador	Adam
	Learning rate	0.001
	betas	(0.9, 0.99)
k (L0 promedio k)		81
	k_aux	512
	alpha_aux	0.03125 (1/32)

7. Loop de Entrenamiento

```

PATIENCE = NAMING_META['extras']['patience']

save_dir = Path('./saved_model')
save_dir.mkdir(exist_ok=True)
best_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],

```

```

        estado='best',
        extras_id=extras_id
    )
BEST_CKPT = save_dir / best_model_name

history = {
    'train_mse_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_mse_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
}

best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'Iniciando entrenamiento: {NUM_EPOCHS} épocas máximo')
print(f'Early stopping: patience={PATIENCE}')
print(f'k = {cfg["k"]} features activas (L0 promedio k por batch)')
print(f'k_aux = {model.k_aux} | alpha_aux = {ALPHA_AUX:.5f} (1/32)')
print(f'Best checkpoint: {BEST_CKPT}')
print('=' * 70)

for epoch in range(NUM_EPOCHS):

    # TRAIN
    model.train()
    t_mse = t_l0 = t_frac = 0.0

    for batch in train_loader:
        x = batch.to(device)
        optimizer.zero_grad()
        recon, hidden, pre_acts = model(x)
        total_loss = loss_function(x, recon, pre_acts, model)
        total_loss.backward()
        model.make_decoder_weights_and_grad_unit_norm()
        optimizer.step()
        model.update_inactive_features(hidden)

    with torch.no_grad():

```

```

        t_mse += F.mse_loss(recon, x).item()
        t_l0 += (hidden > 0).float().sum(dim=1).mean().item()
        t_frac += (hidden > 0).float().mean().item()

n = len(train_loader)
history['train_mse_loss'].append(t_mse / n)
history['train_l0_sparsity'].append(t_l0 / n)
history['train_fraction_active'].append(t_frac / n)

# VALIDATION (sin threshold - igual que training, L0 k)
model.eval()
v_mse = v_l0 = v_frac = 0.0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)
        recon, hidden, _ = model(x)
        v_mse += F.mse_loss(recon, x).item()
        v_l0 += (hidden > 0).float().sum(dim=1).mean().item()
        v_frac += (hidden > 0).float().mean().item()

nv = len(val_loader)
history['val_mse_loss'].append(v_mse / nv)
history['val_l0_sparsity'].append(v_l0 / nv)
history['val_fraction_active'].append(v_frac / nv)

print(f"Época {epoch+1:3d}/{NUM_EPOCHS} | "
      f"Train MSE: {history['train_mse_loss'][-1]:.6f} | "
      f"Val MSE: {history['val_mse_loss'][-1]:.6f} | "
      f"Train L0: {history['train_l0_sparsity'][-1]:.1f} | "
      f"Val L0: {history['val_l0_sparsity'][-1]:.1f}")

# Early stopping
if history['val_mse_loss'][-1] < best_val_mse:
    best_val_mse = history['val_mse_loss'][-1]
    best_epoch = epoch + 1
    epochs_no_improve = 0

torch.save(model.state_dict(), BEST_CKPT)
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],

```

```

        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=cfg,
        history=history,
        optimizer=optimizer
    )
    print(f' Mejor modelo guardado (época {best_epoch} | Val MSE: {best_val_mse:.6f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= PATIENCE:
        print(f'\nEarly stopping en época {epoch+1} (mejor: época {best_epoch})')
        break

# Restaurar mejor modelo
model.load_state_dict(torch.load(BEST_CKPT, weights_only=True))
print(f'\nMejor época: {best_epoch} | Mejor Val MSE: {best_val_mse:.6f}')

# Estimar threshold de inferencia al final del entrenamiento
inference_threshold = model.estimate_threshold(train_loader, device, n_batches=10)
print(f'Threshold de inferencia estimado: {inference_threshold:.4f}')

```

Iniciando entrenamiento: 1000 épocas máximo

Early stopping: patience=20

k = 81 features activas (L0 promedio k por batch)

k_aux = 512 | alpha_aux = 0.03125 (1/32)

Best checkpoint: saved_model\sae_topk_batch-topk_l6_50000g_ex2_best.pt

=====

Época 1/1000 | Train MSE: 0.169227 | Val MSE: 0.137772 | Train L0: 81.0 | Val L0: 81.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt

Mejor modelo guardado (época 1 | Val MSE: 0.137772)

Época 2/1000 | Train MSE: 0.131956 | Val MSE: 0.128504 | Train L0: 81.0 | Val L0: 81.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt

Mejor modelo guardado (época 2 | Val MSE: 0.128504)

Época 3/1000 | Train MSE: 0.126372 | Val MSE: 0.125463 | Train L0: 81.0 | Val L0: 81.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt

Mejor modelo guardado (época 3 | Val MSE: 0.125463)
 Época 4/1000 | Train MSE: 0.124029 | Val MSE: 0.123929 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 4 | Val MSE: 0.123929)
 Época 5/1000 | Train MSE: 0.122733 | Val MSE: 0.123081 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 5 | Val MSE: 0.123081)
 Época 6/1000 | Train MSE: 0.121896 | Val MSE: 0.122495 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 6 | Val MSE: 0.122495)
 Época 7/1000 | Train MSE: 0.121314 | Val MSE: 0.122068 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 7 | Val MSE: 0.122068)
 Época 8/1000 | Train MSE: 0.120880 | Val MSE: 0.121780 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 8 | Val MSE: 0.121780)
 Época 9/1000 | Train MSE: 0.120531 | Val MSE: 0.121595 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 9 | Val MSE: 0.121595)
 Época 10/1000 | Train MSE: 0.120260 | Val MSE: 0.121343 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 10 | Val MSE: 0.121343)
 Época 11/1000 | Train MSE: 0.120042 | Val MSE: 0.121235 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 11 | Val MSE: 0.121235)
 Época 12/1000 | Train MSE: 0.119862 | Val MSE: 0.121115 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 12 | Val MSE: 0.121115)
 Época 13/1000 | Train MSE: 0.119717 | Val MSE: 0.121036 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt
 Mejor modelo guardado (época 13 | Val MSE: 0.121036)
 Época 14/1000 | Train MSE: 0.119598 | Val MSE: 0.120958 | Train L0: 81.0 | Val L0: 81.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex2_best.pt

topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 14 | Val MSE: 0.120958)
Época 15/1000 | Train MSE: 0.119488 | Val MSE: 0.120903 | Train L0: 81.0 | Val L0: 81.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 15 | Val MSE: 0.120903)
Época 16/1000 | Train MSE: 0.119388 | Val MSE: 0.120869 | Train L0: 81.0 | Val L0: 81.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 16 | Val MSE: 0.120869)
Época 17/1000 | Train MSE: 0.119303 | Val MSE: 0.120837 | Train L0: 81.0 | Val L0: 81.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 17 | Val MSE: 0.120837)
Época 18/1000 | Train MSE: 0.119225 | Val MSE: 0.120864 | Train L0: 81.0 | Val L0: 81.0
Época 19/1000 | Train MSE: 0.119155 | Val MSE: 0.120715 | Train L0: 81.0 | Val L0: 81.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 19 | Val MSE: 0.120715)
Época 20/1000 | Train MSE: 0.119086 | Val MSE: 0.120691 | Train L0: 81.0 | Val L0: 81.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 20 | Val MSE: 0.120691)
Época 21/1000 | Train MSE: 0.119029 | Val MSE: 0.120704 | Train L0: 81.0 | Val L0: 81.0
Época 22/1000 | Train MSE: 0.118976 | Val MSE: 0.120644 | Train L0: 81.0 | Val L0: 81.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 22 | Val MSE: 0.120644)
Época 23/1000 | Train MSE: 0.118929 | Val MSE: 0.120651 | Train L0: 81.0 | Val L0: 81.0
Época 24/1000 | Train MSE: 0.118883 | Val MSE: 0.120625 | Train L0: 81.0 | Val L0: 81.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 24 | Val MSE: 0.120625)
Época 25/1000 | Train MSE: 0.118837 | Val MSE: 0.120585 | Train L0: 81.0 | Val L0: 81.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 25 | Val MSE: 0.120585)
Época 26/1000 | Train MSE: 0.118797 | Val MSE: 0.120575 | Train L0: 81.0 | Val L0: 81.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt
Mejor modelo guardado (época 26 | Val MSE: 0.120575)
Época 27/1000 | Train MSE: 0.118762 | Val MSE: 0.120580 | Train L0: 81.0 | Val L0: 81.0
Época 28/1000 | Train MSE: 0.118728 | Val MSE: 0.120558 | Train L0: 81.0 | Val L0: 81.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt

Mejor modelo guardado (época 28 | Val MSE: 0.120558)

Época	29/1000		Train MSE: 0.118700		Val MSE: 0.120563		Train L0: 81.0		Val L0: 81.0
Época	30/1000		Train MSE: 0.118671		Val MSE: 0.120564		Train L0: 81.0		Val L0: 81.0
Época	31/1000		Train MSE: 0.118648		Val MSE: 0.120564		Train L0: 81.0		Val L0: 81.0
Época	32/1000		Train MSE: 0.118621		Val MSE: 0.120730		Train L0: 81.0		Val L0: 81.0
Época	33/1000		Train MSE: 0.118597		Val MSE: 0.120602		Train L0: 81.0		Val L0: 81.0
Época	34/1000		Train MSE: 0.118573		Val MSE: 0.120648		Train L0: 81.0		Val L0: 81.0
Época	35/1000		Train MSE: 0.118550		Val MSE: 0.120546		Train L0: 81.0		Val L0: 81.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt

Mejor modelo guardado (época 35 | Val MSE: 0.120546)

Época	36/1000		Train MSE: 0.118534		Val MSE: 0.120597		Train L0: 81.0		Val L0: 81.0
Época	37/1000		Train MSE: 0.118516		Val MSE: 0.120553		Train L0: 81.0		Val L0: 81.0
Época	38/1000		Train MSE: 0.118504		Val MSE: 0.120532		Train L0: 81.0		Val L0: 81.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex2_best.pt

Mejor modelo guardado (época 38 | Val MSE: 0.120532)

Época	39/1000		Train MSE: 0.118481		Val MSE: 0.120652		Train L0: 81.0		Val L0: 81.0
Época	40/1000		Train MSE: 0.118466		Val MSE: 0.120599		Train L0: 81.0		Val L0: 81.0
Época	41/1000		Train MSE: 0.118452		Val MSE: 0.120572		Train L0: 81.0		Val L0: 81.0
Época	42/1000		Train MSE: 0.118436		Val MSE: 0.120597		Train L0: 81.0		Val L0: 81.0
Época	43/1000		Train MSE: 0.118426		Val MSE: 0.120578		Train L0: 81.0		Val L0: 81.0
Época	44/1000		Train MSE: 0.118412		Val MSE: 0.120604		Train L0: 81.0		Val L0: 81.0
Época	45/1000		Train MSE: 0.118403		Val MSE: 0.120603		Train L0: 81.0		Val L0: 81.0
Época	46/1000		Train MSE: 0.118391		Val MSE: 0.120594		Train L0: 81.0		Val L0: 81.0
Época	47/1000		Train MSE: 0.118384		Val MSE: 0.120655		Train L0: 81.0		Val L0: 81.0
Época	48/1000		Train MSE: 0.118368		Val MSE: 0.120601		Train L0: 81.0		Val L0: 81.0
Época	49/1000		Train MSE: 0.118366		Val MSE: 0.120602		Train L0: 81.0		Val L0: 81.0
Época	50/1000		Train MSE: 0.118353		Val MSE: 0.120638		Train L0: 81.0		Val L0: 81.0
Época	51/1000		Train MSE: 0.118342		Val MSE: 0.120611		Train L0: 81.0		Val L0: 81.0
Época	52/1000		Train MSE: 0.118334		Val MSE: 0.120649		Train L0: 81.0		Val L0: 81.0
Época	53/1000		Train MSE: 0.118325		Val MSE: 0.120614		Train L0: 81.0		Val L0: 81.0
Época	54/1000		Train MSE: 0.118311		Val MSE: 0.120683		Train L0: 81.0		Val L0: 81.0
Época	55/1000		Train MSE: 0.118304		Val MSE: 0.120696		Train L0: 81.0		Val L0: 81.0
Época	56/1000		Train MSE: 0.118292		Val MSE: 0.120760		Train L0: 81.0		Val L0: 81.0
Época	57/1000		Train MSE: 0.118289		Val MSE: 0.120660		Train L0: 81.0		Val L0: 81.0
Época	58/1000		Train MSE: 0.118280		Val MSE: 0.120705		Train L0: 81.0		Val L0: 81.0

Early stopping en época 58 (mejor: época 38)

Mejor época: 38 | Mejor Val MSE: 0.120532

Threshold de inferencia estimado: 0.0021

8. Guardar Modelo

```
# Guardar checkpoint final
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': {
        'architecture': cfg["architecture"],
        'd_in': cfg["d_in"],
        'd_sae': cfg["d_sae"],
        'k': cfg["k"],
        'lr': cfg["lr"],
        'hook_name': cfg["hook_name"],
    },
    'history': history,
}

final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras_id=extras_id
)
checkpoint_path = save_dir / final_model_name
torch.save(checkpoint, checkpoint_path)
print(f'Modelo guardado: {checkpoint_path}')

save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=cfg,
```

```

    history=history,
    optimizer=optimizer
)

```

Modelo guardado: saved_model\sae_topk_batch-topk_l6_50000g_ex2_final.pt

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_batch-topk_l6_50000games\sae_topk_batch-topk_l6_50000g_ex2_final.pt

WindowsPath('C:/Users/Esposa/Documents/Repos/sae-othello-gpt/layer_06/models/sae_topk/sae_topk_batch-topk_l6_50000games/sae_topk_batch-topk_l6_50000g_ex2_final.pt')

9. Métricas básicas del modelo

```

import json

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

training_metrics = {
    # Calidad de reconstrucción
    'val_mse_final': history['val_mse_loss'][-1],
    'val_mse_best': best_val_mse,
    'val_l0_final': history['val_l0_sparsity'][-1],
    'val_fraction_active_final': history['val_fraction_active'][-1],
    # Contexto del entrenamiento
    'best_epoch': best_epoch,
    'total_epochs': len(history['train_mse_loss']),
    'train_mse_final': history['train_mse_loss'][-1],
    'train_l0_final': history['train_l0_sparsity'][-1],
    # Hiperparámetros TopK
    'architecture': cfg["architecture"],
    'k': cfg["k"],
    'd_sae': cfg["d_sae"],
    'd_in': cfg["d_in"],

```

```

    'expansion_factor':      cfg["d_sae"] // cfg["d_in"],
    'hook_name':             cfg["hook_name"],
}

os.makedirs(metrics_dir, exist_ok=True)
metrics_path = Path(metrics_dir) / 'training_metrics.json'
with open(metrics_path, 'w') as f:
    json.dump(training_metrics, f, indent=2)

print(f'Métricas guardadas: {metrics_path}')
print(pd.DataFrame(
    list(training_metrics.items()), columns=['Métrica', 'Valor']
).to_string(index=False))

```

Métricas guardadas: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae\topk_topk\50000games\ex2\metrics\training_metrics.json

Métrica	Valor
val_mse_final	0.120705
val_mse_best	0.120532
val_l0_final	81.0
val_fraction_active_final	0.009888
best_epoch	38
total_epochs	58
train_mse_final	0.11828
train_l0_final	81.0
architecture	batchtopk
k	81
d_sae	8192
d_in	512
expansion_factor	16
hook_name	blocks.5.hook_resid_post

10. Visualización de Resultados

```

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# MSE Loss
axes[0].plot(history['train_mse_loss'], label='Train', color='steelblue', alpha=0.8)
axes[0].plot(history['val_mse_loss'], label='Validación', color='tomato', alpha=0.8)
if best_epoch > 0:

```

```

    axes[0].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7,
                    label=f'Best (época {best_epoch})')
axes[0].set_title('MSE Loss (Reconstrucción)')
axes[0].set_xlabel('Época')
axes[0].set_ylabel('MSE')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

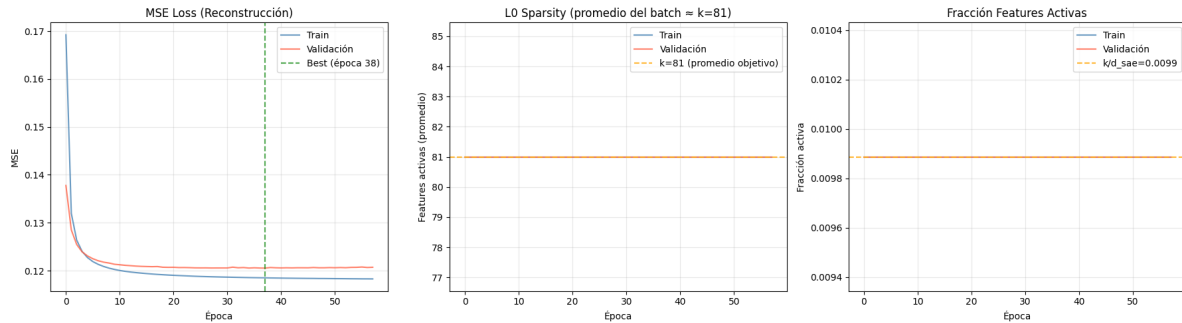
# L0 sparsity (para BatchTopK el promedio del batch k, pero varía por muestra)
axes[1].plot(history['train_l0_sparsity'], label='Train', color='steelblue', alpha=0.8)
axes[1].plot(history['val_l0_sparsity'], label='Validación', color='tomato', alpha=0.8)
axes[1].axhline(cfg["k"], color='orange', linestyle='--', alpha=0.8, label=f'k={cfg["k"]} (p
axes[1].set_title(f'L0 Sparsity (promedio del batch k={cfg["k"]})')
axes[1].set_xlabel('Época')
axes[1].set_ylabel('Features activas (promedio)')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

# Fracción activa
axes[2].plot(history['train_fraction_active'], label='Train', color='steelblue', alpha=0.8)
axes[2].plot(history['val_fraction_active'], label='Validación', color='tomato', alpha=0.8)
expected_frac = cfg["k"] / cfg["d_sae"]
axes[2].axhline(expected_frac, color='orange', linestyle='--', alpha=0.8,
                 label=f'k/d_sae={expected_frac:.4f}')
axes[2].set_title('Fracción Features Activas')
axes[2].set_xlabel('Época')
axes[2].set_ylabel('Fracción activa')
axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(plots_dir / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('Gráficas guardadas exitosamente')
print('='*50)

```



=====

Gráficas guardadas exitosamente

=====

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_batch-topk.ipynb'
```

```

output_path = exp_dir / pdf_name

print(f'Generando PDF con Quarto...')
print(f'Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f'PDF generado exitosamente: {output_path}')
else:
    print(f'Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk_topk\50000games\ex2\sae_topk_batch-topk_16_50000g_ex2_training.pdf

PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk_topk\50000games\ex2\sae_topk_batch-topk_16_50000g_ex2_training.pdf